

Generative AI in Procedural Content Generation for Computer Games: Key Contributions and Trends

Sana Alyaseri

School of Information Technology

Whitecliffe College

Auckland, New Zealand

0009-0004-8650-0109

Abstract—Generative Artificial Intelligence (GenAI) is revolutionizing Procedural Content Generation (PCG) in computer games by enabling the automated creation of diverse, adaptive, and complex game elements. Traditional PCG methods rely on predefined rules and randomness, whereas GenAI introduces machine learning-based approaches that enhance creativity and responsiveness in game design. This paper provides a comprehensive review of GenAI applications in PCG for computer games, discussing its advantages, limitations, and key research challenges. Additionally, we propose future directions for integrating GenAI with human-driven game design to balance automation and artistic control in gaming environments.

Index Terms—Gen AI, Artificial Intelligence, PCG, Computer Games, Procedural content generation

I. INTRODUCTION

PCG has long been integral to game development, facilitating the automated creation of diverse game assets, including levels, textures, characters, and narratives [2]. Traditionally, PCG has relied on algorithmic rules and stochastic processes, which, while effective, often produce repetitive or predictable content. The emergence of GenAI has introduced a transformative shift, offering data-driven approaches that learn from existing game designs to generate novel, high-quality content with minimal human intervention. This shift enhances the scalability and diversity of generated content while also introducing new possibilities for personalized and adaptive gaming experiences.

Since 2020, the integration of GenAI into PCG has accelerated with advancements in deep learning methodologies, including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformers (TF), Diffusion Models (DM), and Reinforcement Learning (RL). These techniques have enabled the creation of more complex and immersive game worlds, expanding the boundaries of game design and interactivity. By leveraging these AI-driven approaches, developers can create dynamic content that adapts to player preferences and behaviors, ultimately enhancing engagement and replayability.

This paper presents a comprehensive narrative review of the role of GenAI in PCG, with a focus on developments since 2020. The key research questions guiding this review are as follows:

- What are the major advancements in GenAI-driven PCG since 2020, and how have they reshaped game development?
- What are the dominant themes and challenges in the application of GenAI to PCG, and how do these vary across different game content types?
- What are the limitations of current approaches, and which future directions offer the most potential for advancing the field?

To systematically address these questions, this review first establishes a foundational understanding of PCG and GenAI, contextualizing recent advancements. It then synthesizes key studies, highlighting notable contributions and emerging trends. Finally, the discussion explores challenges, limitations, and promising future directions, offering insights into how GenAI can further evolve to enhance PCG in games.

II. BACKGROUND

Procedural Content Generation (PCG) involves algorithmically creating game content with minimal user input, either autonomously or alongside human designers. Traditional methods use rules and randomness but often yield repetitive results and require manual tuning, limiting scalability in large-scale games [2]. To overcome these challenges, researchers have increasingly incorporated Artificial Intelligence (AI) techniques into PCG. While AI broadly enables machines to perceive, interpret, act, and learn, a specialised subset known as Generative AI (GenAI) focuses on content creation by learning from vast datasets [5]. GenAI has revolutionised multiple industries, from text generation in ChatGPT-4 to image synthesis in DALL-E. In the context of PCG, it has introduced data-driven approaches that surpass traditional handcrafted rules and stochastic algorithms, enabling richer and more diverse game content [6].

Models such as GAN, VAE, TF, RL, and DM have demonstrated significant success in generating high-quality, immersive game assets with minimal human intervention [7]. By leveraging large datasets, GenAI facilitates the creation of dynamic and interactive content, enhancing player engagement and expanding the creative possibilities of game development. As AI-driven techniques continue to evolve, they offer unprecedented opportunities for scalable and adaptive gaming experiences [8].

III. RESEARCH SCOPE AND SELECTION CRITERIA

This review uses a structured methodology to analyze recent (2020–2025) peer-reviewed literature on GenAI in PCG, focusing on studies directly related to content generation in games. Inclusion and exclusion criteria were applied to exclude peripheral topics like evaluation frameworks or AI-driven behavior modeling.

First, we considered papers published in five academic databases (IEEE Xplore, ACM Digital Library, Scopus, Google Scholar, and conference proceedings). Next, we employed a systematic search methodology to identify relevant studies. We primarily used Google Scholar for academic research and The search queries included:

- (“Procedural Content Generation” OR “PCG”) AND (“Generative AI” OR “AI-generated content”)
- (“Procedural Content Generation” OR “PCG”) AND (“Generative Adversarial Networks” OR “GANs”) AND (“games” OR “game levels”)
- (“Procedural Content Generation” OR “PCG”) AND (“Variational Autoencoders” OR “VAEs”) AND (“games” OR “level generation”)
- (“Procedural Content Generation” OR “PCG”) AND (“Transformers” OR “Deep Learning”) AND (“game AI” OR “game content”)
- (“Procedural Content Generation” OR “PCG”) AND (“Reinforcement Learning” OR “RL”) AND (“game design” OR “level generation”)
- (“Procedural Content Generation” OR “PCG”) AND (“Diffusion Models”) AND (“games” OR “game AI”)

Search strings were tailored for each database, with filters applied to titles, abstracts, and keywords. Results were sorted by relevance and date to highlight recent work. Studies focusing on content generation, such as levels, maps, narratives, and hybrid methods were prioritized

On the other hand, several types of studies were excluded from our review to maintain a focused scope. Specifically, we omitted:

- Studies evaluating (rather than generating) content.
- Works solely on NPC behavior generation.
- Non-game PCG applications (e.g., film, simulations).
- Studies published before 2020.

With the selection process complete, we structured the paper to systematically analyze and present our findings.

IV. RECENT ADVANCES (2020 ONWARDS)

Since 2020, GenAI has rapidly evolved, introducing novel methods PCG in computer games. These advancements have created more sophisticated, diverse, and player-responsive content, transforming how games are designed and experienced. This section explores the key GenAI methods that have driven these innovations, including GAN, VAE, TF, RL and DM, highlighting their applications and impact on game development.

A. Generative Adversarial Networks (GANs)

Recent GAN-based PCG research aims to improve diversity, control, and playability while tackling issues like mode collapse and efficiency. Conditional GANs (CGANs) are used to generate content based on design constraints, for example, to balance level segments or to adjust difficulty by player skill allowing for more precise and designer aligned outputs [10].

Building on this foundation, Adversarial reinforcement learning (ARL) enables dynamic content generation through iterative training between generator and solver agents. Frameworks like ARLPCG [11] and [12] demonstrate high solvability rates, though scalability remains a challenge due to computational demands.

To improve structural coherence, hybrid approaches combine GANs with complementary techniques. For example, Graph+GAN [13] integrates generative graph grammars to produce playable dungeon layouts, while [14] combines VAEs with spatial partitioning for multi-domain level generation. These methods address GAN limitations like fragmented outputs by enforcing domain-specific rules. MultiGAN [15], which trains separate GANs for level segments, significantly enhances playability and diversity. A major challenge remains mode collapse, where GANs generate repetitive content. Recent solutions include WGANs [16] for improved visual diversity, Pix2Pix with L1 regularization [17] for realistic terrain, and MAP-Elites [18], which explores GAN latent spaces to generate novel content aligned with gameplay metrics.

Future research highlights the need for larger training datasets, designer feedback integration, and better evaluation metrics. Advances include CGANs trained on elevation and water data [19], CycleGANs for style transfer [14], and lightweight models for efficiency [10]. Collaborative tools [20] may further align outputs with designer intent. Despite challenges in modeling complex mechanics, these developments position GANs as a flexible foundation for next-gen PCG

[21] introduces MultiGAN, which trains separate GANs on different level segments to improve connectivity and flow in Mega Man levels. Compared to OneGAN, MultiGAN produces longer, more coherent paths with fewer unreachable areas. Human evaluations found MultiGAN levels more fun and human-like. Future directions include using CGANs, larger datasets, and diversity-focused algorithms to enhance cohesion.

In general, recent advances based on GAN-based PCGs such as CGANs, ARL, and hybrid models have improved diversity, control, and playability. These approaches tackle mode collapse and enable more human-like, structurally sound content. Continued progress depends on richer datasets, designer-in-the-loop tools, and better evaluation methods.

B. Variational Autoencoders (VAE)

VAEs are generative models that learn data distributions to create realistic samples, commonly used in image generation and compression [22]. In PCG, they enable diverse content creation, such as levels and avatars. VAEs were combined with quality-diversity methods like MAP-Elites to generate playable

levels in games such as Super Mario Bros., Mega Man, and Castlevania. The Blend-Elites approach enables hybrid-level design, enhancing creative potential [23].

Building on this approach, VAEs enable sequential level generation, creating more coherent and playable levels than segment-based methods. This approach works well for games requiring multi-directional progression (e.g., Mega Man) and allows blending levels from different games (e.g., Super Mario Bros. and Kid Icarus) while maintaining quality in arbitrarily long levels [24].

VAEs also were combined with Example-Driven Binary Space Partitioning (EDBSP) for hybrid-level generation, where VAEs create structural sketches and EDBSP adds details by blending patterns across domains. Tested in seven platformer environments, this approach produces both faithful reproductions and innovative layouts, with potential future extensions including automatic domain grouping and mixed-initiative design tools [14].

Beyond level generation, VAEs demonstrate effectiveness across diverse generative tasks, including game content creation (e.g., Arknights levels), anime-style avatar generation, and MNIST dataset clustering. Their versatility and ease of use suggest promising applications in generative AI, with potential quality improvements through hybrid approaches like VAE-GAN combinations [25].

VAEs show strong potential for game design when combined with QD algorithms, sequential generation, or hybrid approaches like EDBSP. These methods enable diverse, playable content creation, including level blending and property-specific generation. Future directions include full-level generation, user studies on diversity perception, and co-creative tools, along with exploring conditional VAEs and improved domain grouping.

The integration of VAEs into PCG represents a major GenAI advancement. Next steps could combine VAEs with RL for player-responsive generation and apply conditional VAEs across genres, reinforcing their role as foundational models for adaptive game design.

C. Transformers (TFs)

TFs are a powerful deep learning architecture that revolutionized NLP and expanded to computer vision, RL, and scientific computing. Introduced in the seminal "Attention Is All You Need" paper, they underpin models like GPT, BERT, and Vision Transformers [26]. In GenAI and PCG, TFs enable dynamic narrative generation, adaptive game quests, music composition, and coherent procedural content. Their self-attention mechanism and integration with LLMs have significantly advanced quality and diversity in generated game content [27].

The concept of 'AI-Native games' integrates generative AI as a core gameplay element, using LLMs (Chain-of-Thought, ReAct) for narrative coherence and Stable Diffusion with ControlNet for pixel art generation. GPT-4 creates image prompts from story scenes while maintaining character consistency through detailed prompting. Current challenges involve

narrative stability, preventing AI exploitation, and balancing player agency with game structure. Future enhancements could improve combat systems and expand to 3D/animation generation. The gameplay features keyword-based weapon creation and alternating storytelling/combat phases with an AI-driven King [28].

LLMaker is a Windows application using GPT-3.5-turbo-0613 to generate Darkest Dungeon-inspired dungeon crawler levels. It employs context-aware generative grammar and predefined functions to maintain valid level structures within domain constraints. Testing shows function calling produces valid content 100% faster (under 10s response time) compared to error-prone alternatives. Current limitations include potential test bias and GPT's proprietary nature, with future directions exploring open-source models (e.g., NexusRaven-13B) and integrating visual generators like Stable Diffusion [29].

Likewise, NLP models (BERT, GPT-2) can automate context-aware quest generation, addressing the inefficiency of manual creation. Function calling proves most effective, ensuring valid content and fast response times (unaffected by complexity) while avoiding parser/domain errors. Though it sometimes misinterprets inputs. Limitations include potential test bias and reliance on proprietary GPT models, suggesting future exploration of open-source alternatives and integration with visual generators like Stable Diffusion [27].

Additionally, LLMs can democratize PCG by enabling non-technical players to create game content through natural language. The Quick Map Building System (QMBS) demonstrates this by translating player descriptions directly into game elements, highlighting LLMs' potential to bridge language and design. The study stresses the value of integrating players into the creation process itself [30].

Furthermore, A Transformer-based neural network for game-level generation employs embedding layers, positional encoding, and self-attention to produce diverse, structurally coherent levels. Trained on varied datasets, it outperforms traditional PCG methods in coherence, diversity, and engagement across multiple genres (platformers, dungeon crawlers, RTS, etc.). Future directions include multi-modal inputs, real-time adaptation, and tighter game engine integration [31].

Finally, A Transformer model trained on 37 VGLC Super Mario levels achieves 93% tile accuracy and 88.4% playability, surpassing LSTM baselines. It responds to design prompts (blocks, enemies, etc.) while novelty search explores the space without compromising playability. Temperature tuning balances diversity and quality [32]

In conclusion, TF and LLMs in PCG mark a major leap in generative AI for games, enabling diverse, high-quality content like narratives and levels. Combining them with techniques like function calls and self-attention boosts creativity. Future work could improve narrative consistency, explore open-source options, and integrate multi-modal inputs for real-time adaptation, advancing AI in game design.

D. Reinforcement Learning (RL)

RL has emerged as a powerful method for creating adaptive and player-responsive content in games. Unlike other generative methods, RL focuses on learning optimal behaviours through trial and error, making it well-suited for tasks like PCG, enemy AI design, and dynamic difficulty adjustment [6]. The integration of RL into PCG for games has led to significant advancements, as highlighted by several research papers. These studies collectively demonstrate the versatility and potential of RL in generating diverse, high-quality game content, including levels, mechanics, and interactive experiences.

G-PCGRL has been introduced using a reinforcement learning (RL) approach for controllably generating graph-structured data by modifying a graph's adjacency matrix. The method employs a reward function to guide the agent in adding or removing edges that satisfy predefined constraints, ensuring valid graph generation. Compared to random search and evolutionary algorithms, G-PCGRL achieves faster execution times, particularly for complex constraints, and is applied to tasks like game economy and skill tree generation. The authors suggest future extensions, including support for more intricate constraints and experimentation with alternative neural architectures, such as convolutional or graph convolutional layers, to improve scalability [33].

Recent work proposes an RL-based level balancing architecture with a level generator, balancing agent, and simulation-driven rewards. The agent adjusts levels to meet targets (e.g., equal win rates) using rewards from simulations. Swap-based representations (Swap-Narrow, -Turtle, -Wide) improve robustness while preserving playability. Tests show 68% of levels achieve balance, with domain-independent applicability. Future work includes enhancing aesthetics via Stable Diffusion and balancing for varied player types [34].

Similarly, IORand presents a hybrid PCG approach combining RL with semi-random generation for controllable yet novel level creation. The algorithm employs graph-based evaluation of player interactions using metrics like risk, reward, obstruction, precision, motivation, and distance. An RL reward function guides generation toward specific gaming experiences, yielding 64%-76% playable levels with moderate novelty that align well with target designs. Future work includes extending the algorithm's generalization across games and refining experience similarity measures [35].

In addition to level generation, the Experience-Driven RL (EDRL) framework generates endless, playable levels with controlled fun and diversity. It employs experience-driven reward functions evaluating fun, historical deviation, and playability. Results show RL agents produce both playable and engaging levels, with the combined reward function performing best. The scalable framework works for any game with segment-based design and compressed representations. Future research may explore adaptive diversity/surprise and test the framework on more complex games and non-level content [36].

Building on experience-driven RL, another work formulates level generation as a Markov Decision Process (MDP), where agents iteratively modify levels to maximize quality-based rewards. The framework features three core components: problem module, representation module, and change percentage, offering flexibility for human-AI collaboration in mixed-initiative tools. All representation types successfully generated levels, though with varying performance: the Wide representation achieved comparable results with fewer parameters, but occasionally produced lower-quality outputs compared to other representations. Future directions include investigating self-play, hierarchical agents, and better preservation of human edits in collaborative tools [37].

Beyond level generation, a study investigates RL for evaluating and generating game rules in the Mechanic Maker 2.0 environment. A PPO-trained RL agent evaluates rules, while an A* agent uses heuristic pathfinding. The RL agent produces more diverse rules, including unconventional solutions (e.g., wall pass) than A*'s predictable outputs. Results show RL's potential for generating human-friendly, creative rule sets, though it risks fitness function exploitation. Future work includes supporting more variables/effects, conditional rules, and robust fitness functions [38].

In a similar vein, a reinforcement learning-based framework has been developed to generate and evaluate complete game levels, optimizing for both fun and playability. This approach combines level and gameplay rewards to ensure that the generated levels are both functional and enjoyable. Results show that this dual-reward strategy outperforms single-reward setups, achieving high fun scores and near-perfect playability. The framework is also robust and adaptable, capable of generating levels tailored to different playstyles while maintaining high fun scores. This method, known as the Episodic Generative Soft Actor-Critic (EGSAC) framework, was introduced in [39]. Future work includes exploring additional metrics, user testing, and dynamic object placement.

Building on the concept of iterative content generation, a novel framework combines transformer networks with offline reinforcement learning to iteratively generate game levels, modeling content creation as a Markov Decision Process (MDP). It is trained on pre-collected gameplay data using a masked negative log-likelihood loss to preserve causality. When tested on the Sokoban puzzle game, the framework—called PCGPT—produces complex and diverse levels, outperforming PCGRL in solution length, total reward, steps, and structural changes. PCGPT was introduced in [40], with future work aiming to extend it to more complex games and improve evaluation metrics to better reflect player preferences.

An alternative method leverages ARL for PCG, where two agents are trained in opposition: a Generator that creates new environments and a Solver that attempts to solve them. This setup was validated using 3D environments resembling a third-person platformer and a racing game, showing a significantly improved solve ratio. The approach, known as Adversarial RL for PCG (ARLPCG), was proposed in [11], highlighting the effectiveness of adversarial training in producing diverse and

challenging game content.

A framework integrating human expertise with RL agents generates game levels using entropy-based metrics to measure scenario diversity. Agents trained with these rewards produce more playable, diverse levels with minimal repetition. Evaluated in Dungeon, MazeCoin, and Zelda environments, the method demonstrates effective high-quality level generation [41]. Future work includes advanced rule-generation algorithms and refining the fitness function to prevent RL exploitation.

Finally, a RL approach generates high-quality, diverse stages for turn-based RPGs using virtual simulations (VS) to provide intermediate rewards and improve output quality. The results indicate that the RL methods, especially VS-DDPG, produce stages more efficiently than random generation and match supervised learning in quality. The system successfully balances diversity and quality, suiting it for real-time generation. Future work involves adapting the method to complex games and designing evaluation metrics aligned with human preferences [42]

In conclusion, integrating RL into PCG significantly advances GenAI for games. These methods create diverse, high-quality game content, including levels and mechanics. Combining RL with adversarial training, entropy-based diversity metrics, and virtual simulations opens new avenues for engaging game content. Future research could enhance narrative consistency, explore open-source alternatives, integrate multi-modal inputs, and develop real-time adaptation models, further advancing generative AI in game design.

E. Diffusion Models (DM)

DMs are key in generating high-quality images, audio, and 3D assets by refining random noise into coherent data. In game development, models like Stable Diffusion and DALL-E create realistic textures, concept art, and environmental assets, offering diverse, consistent content for prototyping and production. Though promising for real-time applications like dynamic visuals based on player input, their high computational demands limit use in resource-constrained environments. As discussed, the integration of DM into PCG has advanced, enabling the generation of diverse, high-quality game content such as levels, maps, and storyboards.

For example, In Super Mario Bros (SMB), DMs use a UNet architecture with self-attention and temporal embedding to reverse noise-corruption and generate new levels. With one-hot encoding for categorical distributions, the model captures spatial and temporal dependencies. Inspired by Denoising Diffusion Probabilistic Models (DDPM), it adapts training for level generation. The model produces diverse, high-quality levels, nearly indistinguishable from human-designed ones. Performance is evaluated on diversity (edit distance and coverage) and playability (solvable by an A* agent), outperforming MarioGPT in playability (93% vs. 90%). Key innovations include multi-class cross-entropy loss, beta scheduling for noise control, and per-sprite temperature scaling, as detailed in [43].

Similarly, In SMB, DMs are used for level generation, focusing on data representation and model architecture. The levels are represented as categorical distributions via one-hot encoding, and the model is trained on level segments from the Video Game Level Corpus (VGLC). The model, based on a UNet architecture with self-attention and temporal embedding, captures spatial and temporal dependencies. It also incorporates double convolution modules and down/up blocks for feature extraction and reconstruction. The study highlights the potential of DMs to generate diverse, high-quality levels that adhere to design principles and maintain playability, as outlined in [44].

In another application, SketchBoard integrates Stable Diffusion (SD-1.5+CN) for image generation and GPT-3.5 for narrative progression. It transforms basic sketches into detailed storyboard frames and generates corresponding narratives. The system's architecture combines CLIP, U-Net, and VAE for image generation, while GPT-3.5 handles text generation. Fine-tuned with a League of Legends (LoL) dataset, SketchBoard maintains stylistic consistency. Evaluations using Frechet Inception Distance (FID) and CLIP scores show it outperforms baseline models in generating high-quality images with strong semantic alignment to text prompts. User studies indicate high satisfaction, with 92% of participants enjoying the system, as detailed in [45]. Future improvements may focus on customization, real-time collaboration, and support for multiple storyboards.

Furthermore, The "five-dollar model" is a lightweight text-to-image generative architecture for low-dimensional, pixel-size images and categorical samples, generating game maps, sprites, and emoji images from text prompts. Despite its simplicity, it demonstrates strong performance and language understanding. The model uses a pre-trained sentence transformer for text encoding and incorporates noise for output diversity. Evaluated through latent space interpolation, vector arithmetic, and CLIP Score, it shows good semantic understanding and generates aesthetically pleasing images, though it struggles with overfitting in smaller datasets (sprites). Future work may explore additional augmentation strategies and attention mechanisms [46].

Similarly, a diffusion-based framework, trained on a single example level, generates diverse, high-quality game levels of arbitrary sizes, outperforming GAN-based methods by reducing artifacts and improving coherence. Levels are represented using Block2Vec, replacing sparse one-hot encoding with dense semantic embeddings for efficient learning of token relationships. The denoising network modifies the U-Net architecture, removing up/downsampling and attention layers to focus on local patches and avoid overfitting. Tested on Minecraft (3D) and Super Mario Bros (2D) levels, the framework successfully generates diverse and high-quality levels, as detailed in [47].

In contrast, Path of Destruction (PoD) method creates a training dataset by iteratively destroying human-designed levels until they resemble random and noisy levels. A convolutional neural network (CNN) is trained to predict repair

actions based on local observations. Tested on *Zelda*, *Sokoban*, and *Danger Dave*, the PoD method generates playable and unique levels, even with limited training data. It demonstrates strong generalization, producing levels distinct from the training data, particularly excelling at generating levels with local patterns like connected walls due to its focus on local observations [48].

In conclusion, the integration of DMs into PCG marks a significant advancement in generative AI for games, enabling the creation of diverse and high-quality content like levels, maps, and storyboards. Combining DMs with techniques such as self-attention and efficient architectures has opened new possibilities for engaging game content. Future research could focus on improving narrative consistency, exploring open-source alternatives, and developing real-time adaptation models to enhance game design.

V. DISCUSSION, EMERGING TRENDS, AND FUTURE DIRECTIONS

Building upon the recent advances in generative AI for PCG discussed in Section IV, this section explores ongoing challenges, emerging trends, and potential future directions.

Firstly, despite its rapid progress, GenAI still faces several challenges in PCG. Ethical and copyright concerns arise due to the use of AI-generated assets, raising questions about intellectual property rights and content originality. Additionally, computational costs are significant, as training and deploying high-quality generative models require substantial computing resources, limiting accessibility for smaller developers. Furthermore, bias and quality control are critical issues, as AI-generated content may reinforce biases present in training data or produce low-quality, unusable assets without rigorous validation mechanisms. Moreover, player perception and acceptance are crucial, as some players may prefer handcrafted game content over AI-generated assets, necessitating a balance between automation and human creativity. Recent studies highlight mixed player reception of AI-generated content. For instance, [42] found that stages generated via RL achieved comparable engagement to handcrafted designs, while [32] reported player preferences for human-like level coherence in TF generated content. Therefore, further research such as longitudinal user studies comparing AI and human content across genres is critical to understanding acceptance barriers.

Moving on to the emerging trends, we analyze how different generative AI techniques have evolved. Table I presents a breakdown of research papers published over recent years, categorized by method.

Based on Table I, we observe that GANs dominated early research (2020–2021) but declined in popularity by 2023. In contrast, VAEs gained traction in 2023 for hybrid and sequential generation tasks. Meanwhile, TF surged in 2024, driven by advancements in LLMs for narrative and level generation. RL has seen consistent use across all years, primarily for adaptive and player-responsive content generation. Additionally, DMs gained prominence in 2023–2024, particularly for high-quality level and storyboard generation.

TABLE I
DISTRIBUTION OF PAPERS BY YEAR AND METHOD

Year	GANs	VAEs	TF	RL	DM	Total Papers
2020	4	0	0	2	0	6
2021	4	0	0	2	0	6
2022	0	0	0	1	1	2
2023	2	4	1	3	3	13
2024	0	0	5	0	2	7
2025	0	0	0	1	1	2
Total	10	4	6	9	7	36

As we look towards the future, several key directions are likely to shape the future of PCG research. Hybrid AI-human collaboration is one such direction, enhancing AI-assisted content creation by integrating human expertise into generative workflows. Improved AI interpretability is another critical area, focusing on developing transparent AI models that enable game designers to better understand and control generative outputs. Furthermore, real-time generative capabilities are advancing, aiming to provide on-the-fly generation techniques for dynamic, personalized game experiences.

Initially, GANs gained traction as a breakthrough in generative AI. However, the rise of DMs and TF introduced significant improvements, making these newer approaches preferred alternatives. Compared to GANs, DMs offer more stable training, fewer convergence issues, and superior output quality. Meanwhile, TF based models excel in natural language interaction, enabling intuitive AI-driven content generation. This shift underscores the importance of adaptability in adopting state-of-the-art generative techniques.

A recurring theme in PCG research is the integration of multiple generative approaches. For example, standard GANs often fail to achieve precise control over output characteristics. cGANs address this by refining content generation, yet they still require further optimization. To improve results, researchers have applied optimization techniques to the latent space, guiding model outputs toward desired properties [49]. This highlights how combining different AI techniques can enhance generative outcomes.

Additionally, it is essential to distinguish GenAI from RL. GenAI typically relies on supervised or unsupervised learning, deriving patterns from large datasets. In contrast, RL follows predefined rules and learns through interaction with an environment. In PCG applications, when vast amounts of training data are available, GenAI is often the optimal choice. However, for cases with limited data or dynamically evolving rules, RL becomes a more suitable alternative. Understanding these distinctions helps in selecting the most effective approach based on specific game design requirements.

A new trend in PCG research involves merging different AI paradigms especially GANs, RL, and LLMs—to create more advanced and context-aware generative systems. This hybridization approach has the potential to redefine the future of PCG in games.

Finally, Table II provides a comparative overview of GenAI

methods discussed in Section IV, summarizing their strengths, weaknesses, and primary applications.

VI. CONCLUSION

The advances in GenAI since 2020 have fundamentally reshaped the landscape of PCG, equipping game developers with innovative tools to create sophisticated, diverse, and player-responsive content. As demonstrated in this review, methods such as GAN, VAE, TF, RL, and DMs offer unique strengths, from fine-grained controllability to context-aware narratives while also grappling with distinct challenges. These include mode collapse in GANs, training inefficiency in RL, and computational demands in DM. Addressing these limitations is critical to unlocking the full potential of GenAI, particularly as hybrid frameworks emerge to balance automation with creativity, such as RL-augmented VAEs or LLM-guided diffusion pipelines.

Ethical considerations, such as copyright risks and bias in training data, alongside technical barriers such as resource accessibility for smaller studios, further underscore the need for responsible adoption. By integrating human-AI collaboration, where designers refine AI output for pacing, novelty, and player engagement, the industry can mitigate these challenges while fostering innovation. The evolution of GenAI promises not only to redefine game design workflows, but also to democratize content creation, enabling dynamic, personalized experiences at scale. Future efforts must prioritize lightweight architectures, ethical guidelines, and player-centric validation to ensure that these tools evolve as collaborative partners rather than as mere automation engines. In doing so, the gaming industry stands poised to enter an era of unprecedented creativity, where human ingenuity and AI-driven scalability co-exist synergistically.

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Method	Strengths	Weaknesses	Primary Use Cases
GANs	High diversity, fine-grained control	Mode collapse, computational cost	Level generation, texture blending
VAEs	Structural coherence, hybrid blending	Limited output quality without hybrids	Sequential levels, cross-game blending
TF	Context-aware narratives, language interaction	High data requirements, bias risks	Quests, dialogue, player-driven stories
RL	Adaptive, player-responsive content	Training inefficiency, reward design complexity	Dynamic difficulty, playtesting
DMs	Photorealistic assets, stability in training	Resource-intensive, slow inference	Textures, 3D assets, storyboards

TABLE II
COMPARISON OF AI METHODS FOR PROCEDURAL CONTENT GENERATION

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